Lead storage battery

Anode : spongy lead

Cathode : lead plate bearing $PbO_2$

Electrolyte : 38% by mass of $H_2SO_4$ with density 1.2g / mL.

Oxidation occurs at the anode

$$Pb(s) \rightarrow Pb^{2+}(aq)+2e^- \quad .....(1)$$

The $Pb^{2+}$ ions combine with $SO_4^{-2}$ to form $PbSO_4$ precipitate.

$$Pb^{2+}(aq) + SO_4^{2-}(aq) \rightarrow PbSO_4\ (s) \quad .....(2)$$

Reduction occurs at the cathode

$$PbO_2\ (s) + 4\ H^+(aq) + 2e^- \rightarrow Pb^{2+}\ (aq) + 2H_2O\ (l) \quad .....(3)$$

The $Pb^{2+}$ ions also combine with $SO_4^{2-}$ ions from sulphuric acid to form $PbSO_4$ precipitate.

$$Pb^{2+}\ (aq) + SO_4^{2-}(aq) \rightarrow PbSO_4\,(s) \quad .....(4)$$

The Overall reactions is

Equation (1) + (2) + (3) + (4)

$$Pb\ (s) + PbO_2\ (s) + 4H^+(aq) + 2SO_4^{2-}\ (aq) \rightarrow 2\ PbSO_4\ (s) + 2H_2O\ (l)$$

The emf of a single cell is about 2V . Usually six such cells are combined in series to produce 12volt

The emf of the cell depends on the concentration of $H_2SO_4$. As the cell reaction uses $SO_4^{2-}$ ions, the concentration $H_2SO_4$ decreases. When the cell potential falls to about 1.8V, the cell has to be recharged.

## Recharge of the cell

As said earlier, a potential greater than 2V is applied across the electrodes, the cell reactions that take place during the discharge process are reversed. During recharge process, the role of anode and cathode is reversed and $H_2SO_4$ is regenerated.

Oxidation occurs at the cathode ( now act as anode)

$$\overset{+2}{Pb}SO_4\ (s) + 2H_2O\ (l) \rightarrow \overset{+4}{Pb}O_2\ (s) + 4\ H^+(aq) + SO_4^{2-}\ (aq) + 2e^-$$

Reduction occurs at the anode (now act as cathode) $PbSO_4(s) + 2e^- \rightarrow Pb(s) + SO_4^{2-}(aq)$

Overall reaction

$$2PbSO_4\ (s) + 2H_2O\ (l) \rightarrow Pb\ (s) + PbO_2\ (s) + 4H^+(aq) + 2SO_4^{2-}\ (aq).$$

Thus, the overall cell reaction is exactly the reverse of the redox reaction which takes place while discharging .

## Uses:

Used in automobiles, trains, inverters etc…

## The lithium – ion Battery

Anode : Porus graphite

Cathode : transition metal oxide such as $CoO_2$.

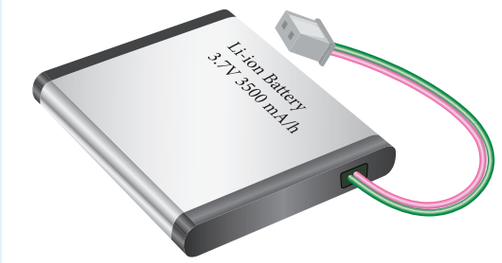



**Figure 9.12 lithium – ion Battery**



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